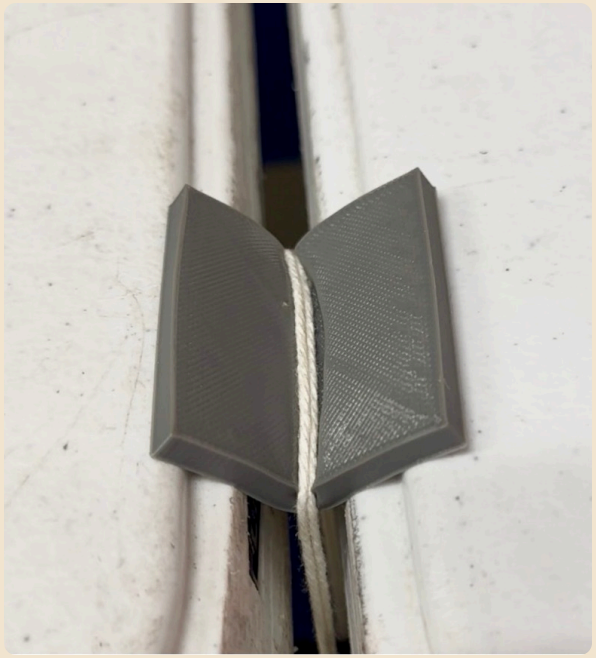
VALUE STATEMENT

## WHAT ORTHOPRINT PROVIDES

**OrthoPrint gives an injured astronaut a splint made for their own ankle, printed on the lunar base when it is needed. The crew gets real support without launching a box of splints in every size.**

| | |
|---|---|
| **Support** | Holds the ankle still so a torn ligament can heal. |
| **A true fit** | Made from a scan of the patient's own leg. |
| **Independence** | Straps the astronaut can tighten with one hand. |
| **Clean wear** | Comes off for washing and skin checks. |
| **Less launch mass** | Print one splint instead of stocking every size. |



Our printed prototype plates during the hanging-weight test

**See the full CAD online**
gertrudish978-pixel.github.io/lunar-splint

YOUR DESIGN · OTHER DESIGN FEATURES

## SPECS & MATERIALS

| ORTHOPRINT | PROTOTYPE #1 · OCT 2026 |
|---|---|
| INJURY | Lateral ankle ligament tear |
| FIT | Custom, from a limb scan |
| CLOSURE | Reusable zip-tie style straps, X pattern |
| SHELL | Solid wall; small vents being re-tested |
| PRINTER | Habitat FDM or SLA |
| SIZE | 460.59 mm tall; plates 144 mm apart |
| MASS | To be measured from the final CAD |
| PRINT TIME | To be measured from the final CAD |

**PLA**
Prototype 1

**ABS**
Prototype 2

**TPU**
Prototype 3 · flexes, firm when solid

**PETG**
Tested · broke when bent

**CF-PEEK**
The material we hope to use for flight



Printed sample showing the shell thickness

### NEXT STEPS

→ Test strap thickness and the load at which straps break
→ Find the temperature where the filament softens
→ Prototype #2 for the Critical Design Review

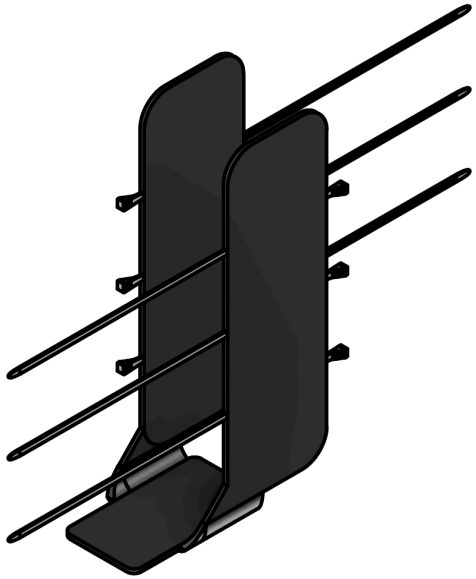
A NASA HUNCH 2026–27 student design project. Not a medical device.


**GLADYS PORTER EARLY COLLEGE HIGH SCHOOL**
BROWNSVILLE, TX · NASA HUNCH 2026–27


NASA HUNCH · LUNAR SPLINT DESIGN CHALLENGE

# ORTHOPRINT

**PDR** Preliminary Design Review · Oct 21–22, 2026



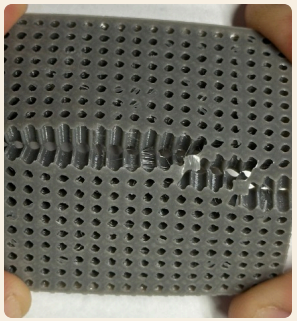
| | |
|---|---|
| NASA PROJECT | Lunar Splint Design Challenge |
| PROGRAM | Design & Prototype |
| SCHOOL | Gladys Porter Early College High School |
| LOCATION | Brownsville, TX |
| TEACHERS | Lupita C. Vasquez and Claudia Cortez |
| TEAM LUNOVA | Gertrudis, Jose and Nancy |

**Website, videos and CAD**
Scan for our testing and prototype videos, CAD and engineering notebook
gertrudish978-pixel.github.io/lunar-splint

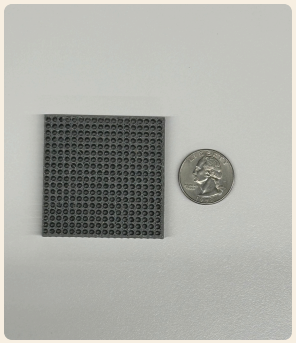
PROTOTYPE HIGHLIGHTS

## FROM SCAN TO SUPPORT

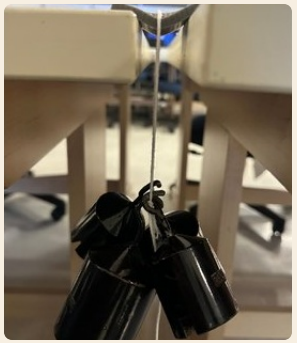
1. **Scan**
   Scan the injured astronaut's lower leg.
2. **Print**
   Print the custom shell and straps on the habitat printer.
3. **Fit**
   Unlatch the splint and fit it on vertically.
4. **Secure**
   Wrap the straps in an X, three times around the ankle.



Sample with holes split, so the wall went solid



TPU sample, quarter for scale



Hanging weights on a sample



Solid TPU, the wall we chose

### TEST RESULTS

| | |
|---|---|
| **PETG, holes:** broke after bending | REJECTED |
| **TPU, holes:** layers split | REJECTED |
| **Solid TPU:** firm, moulds slightly | CHOSEN |
| **Straps at key points:** ankle stayed still | CHOSEN |

YOUR DESIGN · UNIQUE ASPECTS

## BUILT FOR THE MOON

INNOVATION 1

### SINGLE-PIECE RATCHET CLOSURE

Provides a secure hold the astronaut can tighten or release alone, with no loose parts to float away.

INNOVATION 2

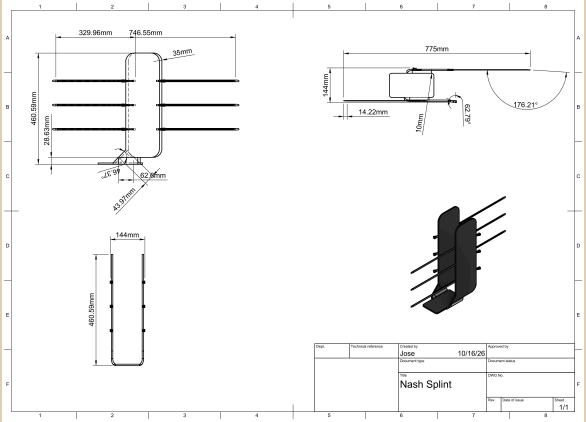
### INTEGRATED HYGIENE

Provides a splint that stays clean and lets the skin be checked over weeks of wear.

INNOVATION 3

### THERMAL PHASE DESIGN

Provides a shell flexible enough to put on and firm enough to hold the ankle.




Detail drawing, dimensions in mm · Jose, 10/16/2026

YOUR DESIGN · BEYOND THE REQUIREMENTS

## MORE THAN THE CHALLENGE ASKED FOR

1. **Three lunar innovations**
   The challenge asks for at least one.
2. **Custom fit from a scan**
   Each patient gets their own splint instead of the closest fixed size.
3. **One-handed use**
   The injured astronaut can tighten the straps alone.
4. **Backed by user research**
   A Needs Assessment Survey of 56 students and an interview with our school's athletic trainer.
5. **Four materials tested**
   PLA, ABS, TPU and PETG, with CF-PEEK recommended for flight.

### NEEDS ASSESSMENT SURVEY · 56 STUDENTS

| | |
|---|---|
| Ankle injured most | 63% |
| Can't find a good fit | 54% |
| On their feet a lot | 96% |
| Devices help healing | 98% |

### WHY THE MOON NEEDS IT

A return to Earth takes days, so the crew needs a splint they can make on site.

| | | |
|---|---|---|
| **$1M** launch cost per kilogram | **1/6 g** lunar gravity | **−1–2%** bone density per month |